CSC 446 Course Project

“Simulation of Emergency Department Operations Using Discrete Event Simulation”

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Problem Description

Emergency departments (EDs) often face rising patient volumes and heterogeneous case complexities, resulting in overcrowding, prolonged waiting times, and challenges in resource allocation. The Manchester Triage Protocol classifies patients into priority levels (Red, Orange, Yellow, Green, and Blue), each with recommended maximum waiting times. Maintaining compliance with these standards requires efficient resource distribution, particularly of doctors, nurses, and treatment beds.

The primary goal of this simulation study is to quantitatively assess how varying ED resource configurations influence key performance indicators (KPIs) such as patient waiting times, total time in system, and resource utilization rates. By exploring multiple scenarios—adjusting treatment capacities and examining different patient arrival patterns—this study aims to provide actionable insights for optimizing staffing, managing patient flow, and improving overall ED efficiency.

Simulation Model

This simulation study is built upon a **Priority Queueing Model**, which effectively maps the patient flow and resource allocation within an emergency department (ED). Based on the Manchester Triage Protocol, patients are categorized into priority levels that reflect the severity of their medical conditions and the target waiting times for receiving treatment. The model dynamically simulates patient arrivals, triage procedures, and treatment processes to evaluate the performance of the emergency department under various scenarios.

The simulation model is composed of three key stages: **patient arrival**, **triage**, and **treatment**. In the **patient arrival stage**, a Poisson arrival process is used to model the random nature of patient arrivals. The inter-arrival times follow an exponential distribution, ensuring the stochastic characteristics align with real-world hospital data. Upon arrival, each patient proceeds to the **triage stage**, where they are assigned a priority level: Red (immediate), Orange (very urgent), Yellow (urgent), Green (standard), or Blue (non-urgent). This priority assignment determines the waiting time thresholds and resource allocation priorities, reflecting the real-world operational policies of the emergency department.

In the **treatment stage**, patients are directed to designated treatment areas based on their assigned priority levels. The treatment areas consist of two types of critical resources: doctors and treatment beds. Resource allocation follows a priority-based strategy, where patients with higher urgency levels (e.g., Red or Orange) are prioritized over lower-priority patients (e.g., Green or Blue). To closely simulate the treatment process, treatment time is divided into three phases: the initial observation, the intermediate treatment phase, and a final observation. This breakdown ensures the simulation reflects the actual workflow observed in emergency departments.

The following diagram illustrates the patient flow model, including the transition from arrival, triage, and treatment phases:

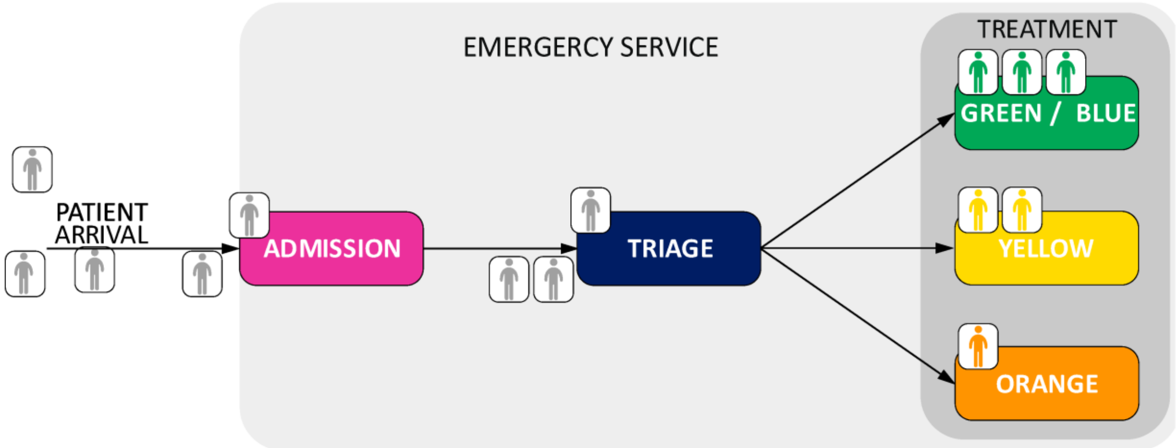


Figure 1: Flowchart representing the patient flow from arrival to treatment zones in the simulation model.

In summary, this priority queueing model integrates stochastic patient arrivals, priority-based resource allocation, and treatment workflows to analyze key performance indicators such as waiting times, system congestion, and resource utilization. This structured framework provides a robust foundation for evaluating and improving the operational efficiency of emergency departments.

Simulation Parameters

In this simulation study, the parameters were carefully selected to ensure that the model accurately reflects the operational behavior of a real-world emergency department (ED). The parameters include patient arrival rates, triage probabilities, resource capacities, and treatment time distributions. These settings are derived from both theoretical distributions and insights from previous studies on emergency department workflows.

The **patient arrival rates** were modeled using a Poisson arrival process, with arrival rates varying by the hour and day of the week. This assumption ensures that the randomness of patient arrivals is captured, as it closely aligns with the stochastic nature of real-life EDs. The weekly arrival rates were scaled based on typical emergency room data, ensuring that the simulation generates realistic patient flows over the course of a simulated week.

The **triage probabilities** are based on the Manchester Triage Protocol and reflect the proportion of patients categorized into each priority level. Specifically, the probabilities are set as follows: Red (0.2%), Orange (8.2%), Yellow (46.9%), Green (41.2%), and Blue (3.5%). These probabilities align with the severity distributions observed in emergency departments, where a large proportion of patients fall under non-urgent (Green) and semi-urgent (Yellow) categories, while immediate emergencies (Red) are relatively rare.

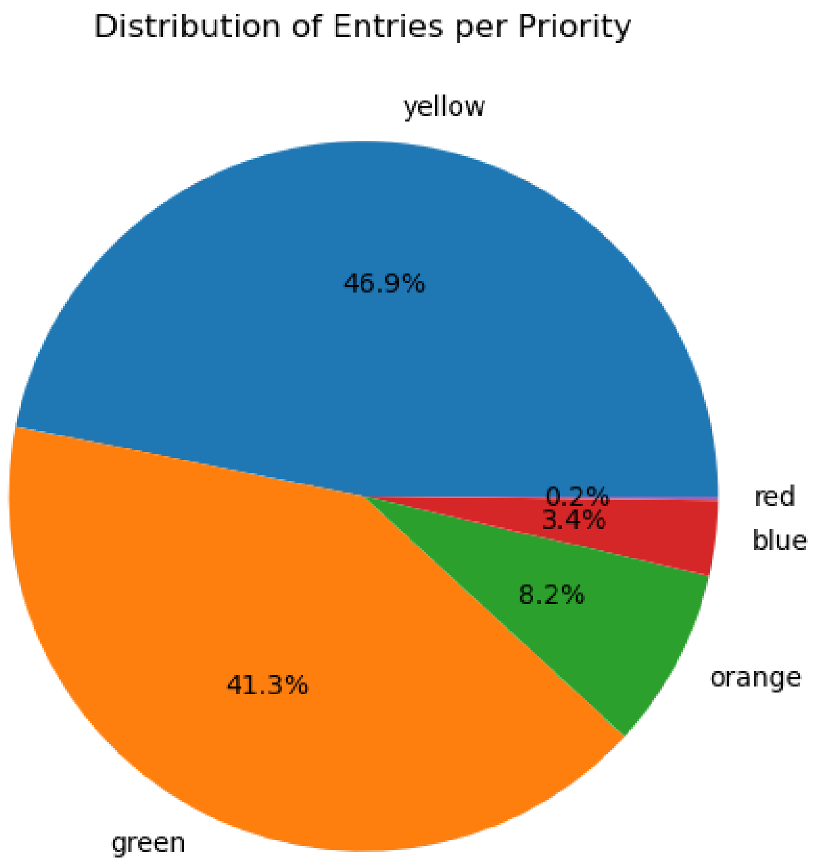


Figure 2: Pie chart showing the distribution of entries per priority level in the simulation model.

The **resource capacities** for admission staff, triage nurses, doctors, and treatment beds were configured to match realistic staffing levels found in emergency departments. For example, during the daytime shift (9 AM to 9 PM), the model assumes 2 admission staff, 3 triage nurses, 2 doctors for Green/Blue patients, 3 doctors for Yellow patients, and 2 doctors for Orange patients. At night (9 PM to 9 AM), the number of staff is reduced to account for lower patient demand, reflecting common hospital operational policies. Additionally, the bed capacities are set to 22 for Green/Blue, 15 for Yellow, and 20 for Orange, which ensures that resource bottlenecks can be evaluated under different scenarios.

The **treatment times** are modeled using a log-normal distribution, based on prior studies that analyzed emergency department data. The treatment time is divided into three distinct phases: an initial observation, a middle treatment phase, and a final observation, representing the typical workflow of patient care in an ED. This breakdown allows the model to capture both the complexity of the treatment process and the varying resource usage across time.

By carefully configuring these parameters, the simulation is able to replicate real-world emergency department operations. This parameter setup allows us to analyze key performance metrics such as average waiting times, resource utilization rates, and system congestion. Furthermore, the chosen settings facilitate scenario testing, enabling the evaluation of system behavior under different capacities, staffing levels, and patient arrival patterns. This ensures that the simulation results are both realistic and actionable, providing insights for optimizing ED performance.

Methodology

This section details the methodology followed to build the simulation, set up the parameters, and collect the statistical results for analysis.

The simulation was implemented using **SimPy**, a Python-based discrete-event simulation library. The overall workflow mirrors the actual operation of an emergency department (ED), capturing the key processes such as patient arrival, triage, and treatment. The simulation begins with **patient arrivals**, modeled using a Poisson process where inter-arrival times follow an **exponential distribution**. This probabilistic approach ensures realistic randomness and variability in patient inflow.

Upon arrival, each patient undergoes an **admission process** that models administrative tasks like registration. The duration of admission is drawn from a **uniform distribution** ranging between 3 and 5 minutes, introducing slight variability without significant complexity. After admission, patients proceed to the **triage process**, where they are assessed and assigned a priority level based on the **Manchester Triage Protocol**. Patients are categorized into five levels of severity: **Red** (immediate care), **Orange** (very urgent care), **Yellow** (urgent care), **Green** (standard care), and **Blue** (non-urgent care).

Once triage is complete, patients enter the **treatment stage**, where they are directed to designated treatment zones according to their priority. Resource constraints such as limited doctors and treatment beds are modeled for each zone. Treatment time is divided into three phases to simulate real-world workflows: an initial observation (15-20% of total time), an intermediate treatment phase, and a final follow-up (5% of total time). Treatment durations are drawn from **log-normal distributions**, reflecting realistic variability in the time required to care for patients of differing severity.

The model also incorporates **shift schedules** to manage resource availability. Day shifts and night shifts were simulated with varying staff capacities, accounting for reduced resources during off-peak hours. Resource allocation follows a **priority queueing mechanism**, where patients with higher urgency (e.g., Orange) are prioritized over those with lower urgency (e.g., Green/Blue).

The simulation parameters were carefully designed to reflect realistic emergency department operations and ensure the validity of the model. **Patient arrival rates** were modeled as hourly variations for each day of the week. This approach accounts for temporal patterns observed in real emergency departments, where patient arrivals peak during certain hours and drop during off-peak periods. The Poisson process was chosen because it is widely used to represent random arrivals in queuing systems. **Triage times** were fitted to a **log-normal distribution** based on realistic observations, as this distribution accommodates skewed data where most cases are quick but outliers take longer. Similarly, treatment times for patients of different priority levels were assigned log-normal distributions, calibrated as follows: **Orange** patients have shorter treatment durations due to urgency, **Yellow** patients require intermediate treatment time, and **Green/Blue** patients typically experience longer treatment durations.

Resource settings included the number of available doctors and treatment beds for each zone (Green/Blue, Yellow, and Orange). These values were derived from observed staff levels and treatment capacities in real-world hospitals. The Manchester Triage Protocol's waiting time thresholds were also incorporated: **Green/Blue** (240 minutes), **Yellow** (60 minutes), and **Orange** (10 minutes). Two shifts were simulated to reflect actual ED operations: a day shift (09:00-21:00) and a night shift (21:00-09:00). Staffing levels during night shifts were reduced to reflect lower patient demand and resource availability. These parameter settings ensure that the simulation captures the variability and constraints of ED operations while aligning with observed data and protocols.

To collect reliable statistics, the simulation was executed for **7 consecutive days** (one week of ED operations) with multiple independent runs. Each run was initialized with a different random seed to account for stochastic variability and ensure robustness. For each patient, the following key performance metrics were recorded: **arrival time**, **waiting time**, and **total time in system**. Resource usage data were collected for both doctors and treatment beds.

Utilization rates were calculated as the percentage of time a resource was actively in use during the simulation. This metric helps assess system congestion and identify underutilized or overloaded resources.

Multiple scenarios were simulated to evaluate the impact of resource capacities on waiting times and resource utilization. For example, the number of treatment beds was varied in the Green/Blue, Yellow, and Orange zones. For each scenario, statistics were aggregated over **five independent runs** to account for randomness. The average values of collected metrics were calculated, and **95% confidence intervals (CIs)** were derived using the **t-distribution**. This ensures statistical robustness and allows comparisons across scenarios.

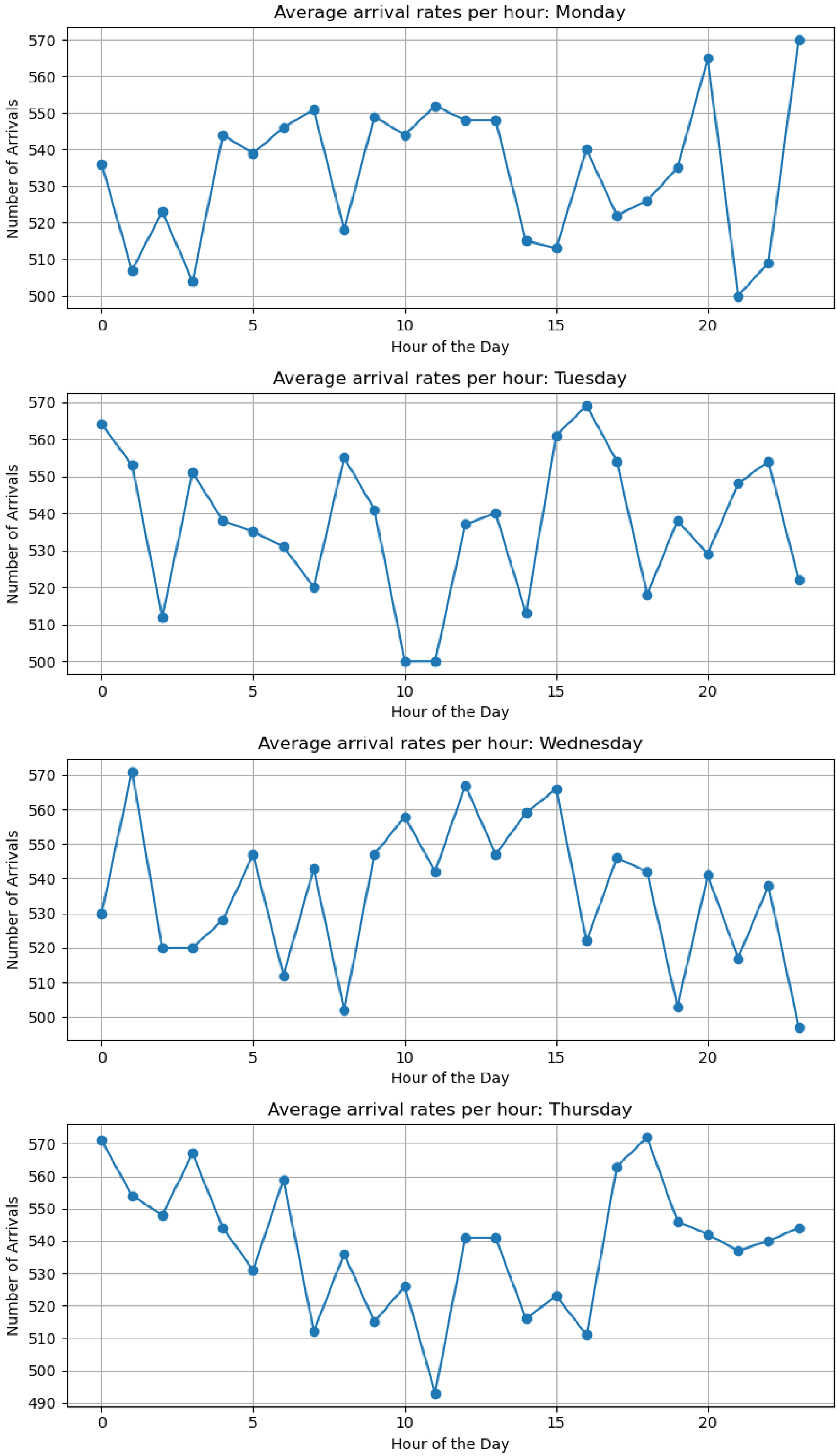
The collected statistics provide a comprehensive understanding of the ED’s performance, including patient waiting times, resource utilization rates, and the system’s ability to meet Manchester Triage Protocol thresholds under varying conditions.

Results & Analysis

This section presents a comprehensive analysis of the simulation results. We begin by examining arrival rates and admission times, followed by fitting distributions for triage and treatment times. We then explore scenario analyses for three different priority zones (Green, Yellow, and Orange) by varying resource capacities. All scenario analyses are based on at least five independent runs (each with a different random seed) to ensure statistical validity. The reported metrics include mean waiting times, treatment times, and occupancy rates, along with 95% confidence intervals (CIs).

Arrival Rates and Preliminary Observations

In examining patient arrival patterns, the simulation provided data on hourly arrival rates for each weekday. Aggregating data from multiple runs ensured stable estimates. The following figure illustrates the average hourly arrival rates from Monday through Sunday, highlighting peak load hours. These insights can guide resource allocation, suggesting increased staffing during peak hours and cost-saving measures during low-demand periods without compromising care quality.


Average arrival rates per hour: Monday
Number of Arrivals
Hour of the Day
Average arrival rates per hour: Tuesday
Number of Arrivals
Hour of the Day
Average arrival rates per hour: Wednesday
Number of Arrivals
Hour of the Day
Average arrival rates per hour: Thursday
Number of Arrivals
Hour of the Day

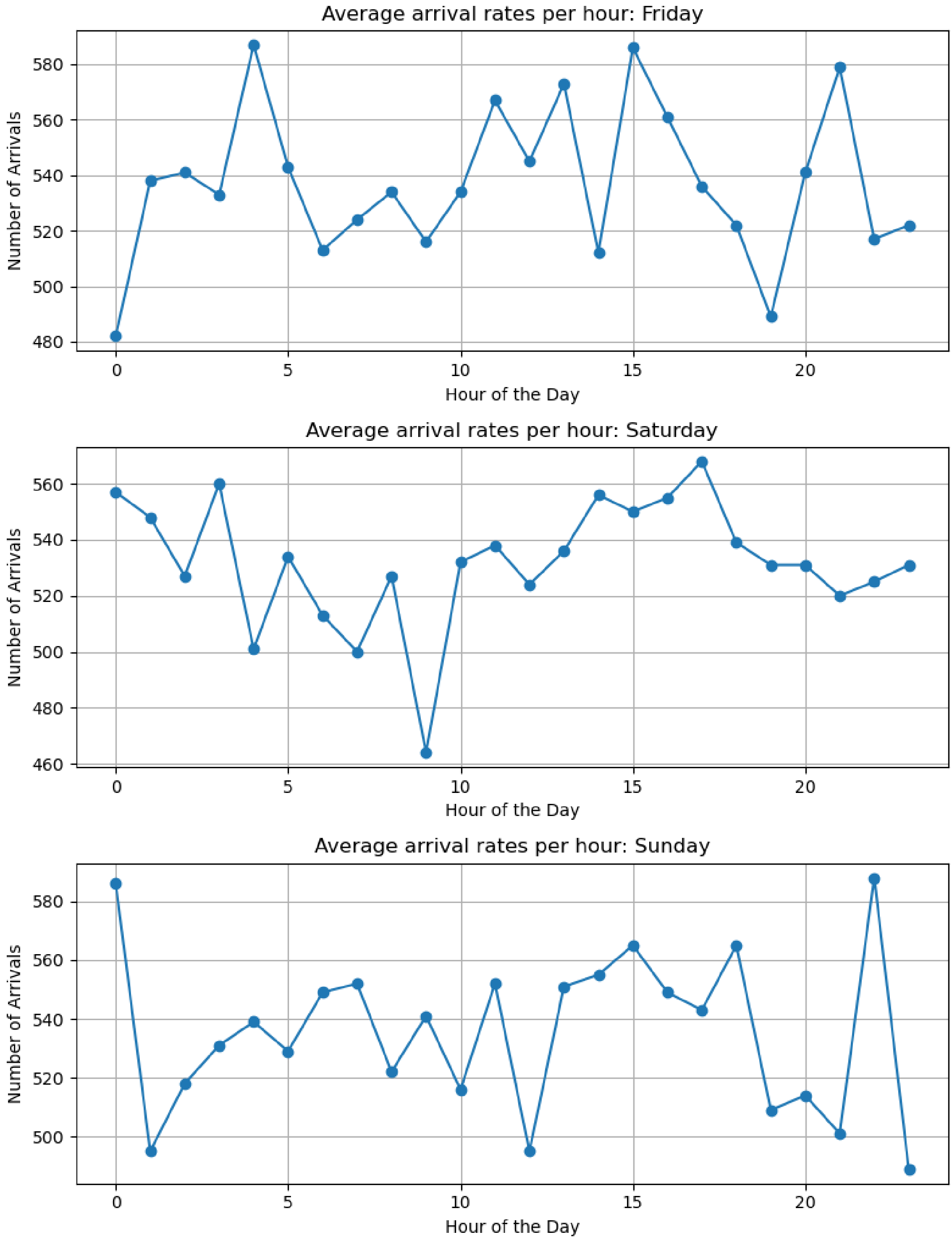


Figure 3: Hourly arrival rates for each day of the week. Each point represents the aggregated average from multiple runs, ensuring stable estimates and revealing peak load hours.

Admission Times

Admission times represent the first step in the patient journey. Analysis of the aggregated admission times revealed a uniform distribution between 3 and 5 minutes, as depicted in **Figure 4**. This consistency indicates a stable admission process that does not pose a significant bottleneck.

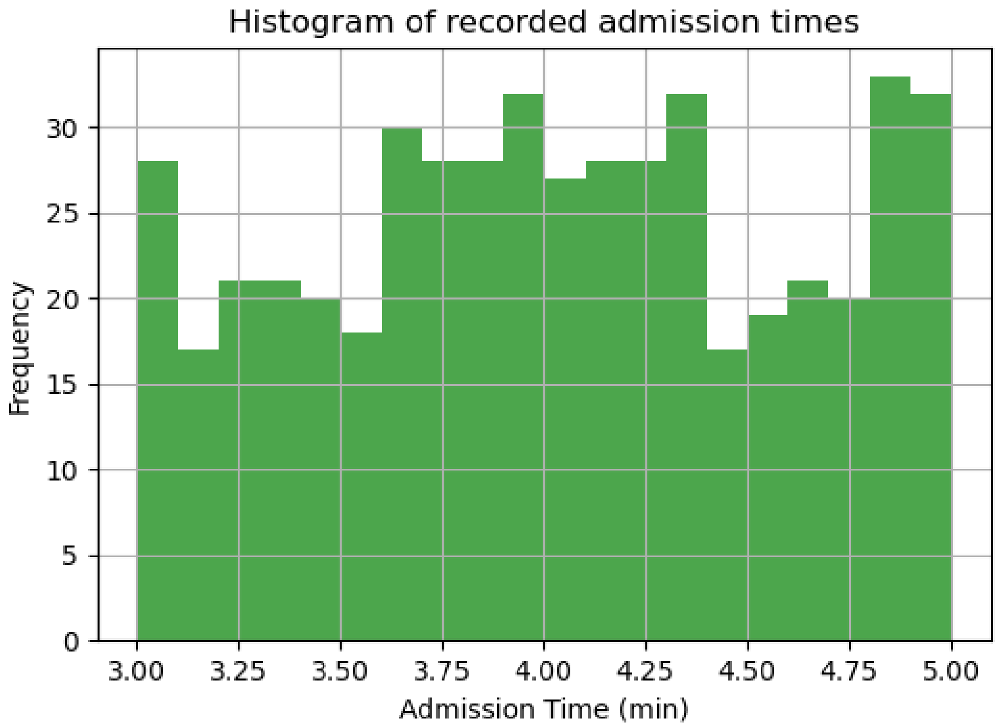


Figure 4: Admission time distribution across multiple runs. The uniform spread suggests a consistent admission workflow.

Triage Times

Triage times, which determine patient priority, were analyzed by compiling data from multiple runs. A lognormal distribution was fitted to the observed times. **Figure 5** illustrates the histogram and fitted curve, along with a probability plot to validate the fit. This analysis confirms the suitability of the lognormal distribution for modeling triage times.

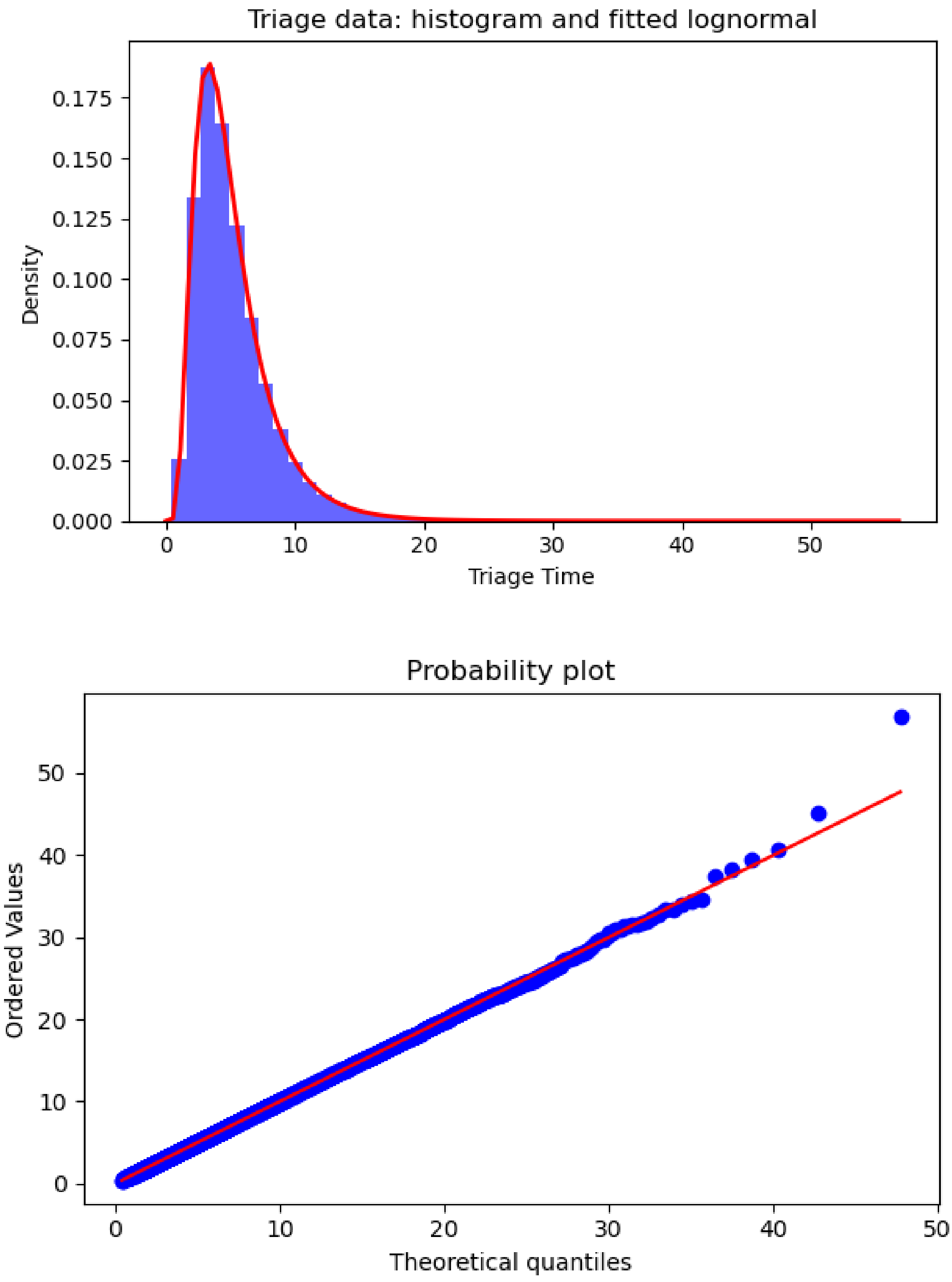
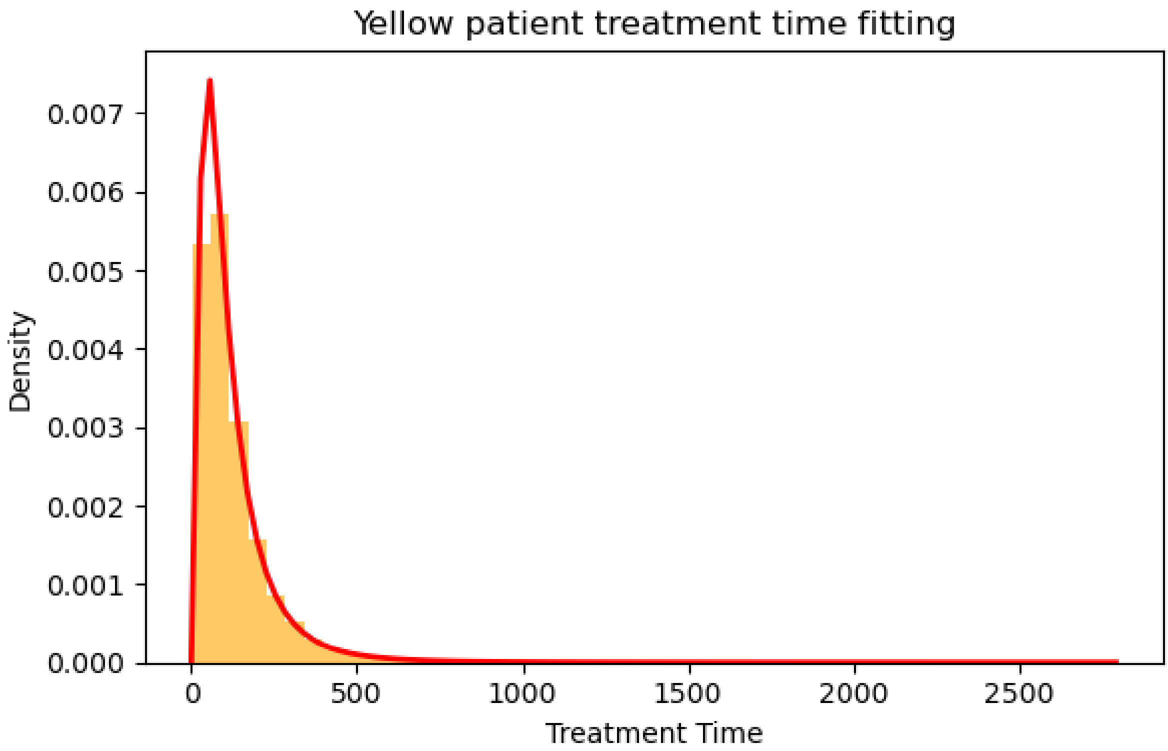


Figure 5: Triage time distribution and probability plot. Most observations fit a lognormal pattern, with a few outliers at longer durations. Data are aggregated over multiple runs, ensuring robust parameter estimation.

Treatment Times for Yellow Priority Patients

Treatment times, especially for Yellow priority patients, were also modeled using a lognormal distribution. The following presents the histogram and the fitted curve, demonstrating that most treatment durations are short, with occasional outliers representing complex cases. These findings underscore the importance of sufficient capacity to handle such cases.



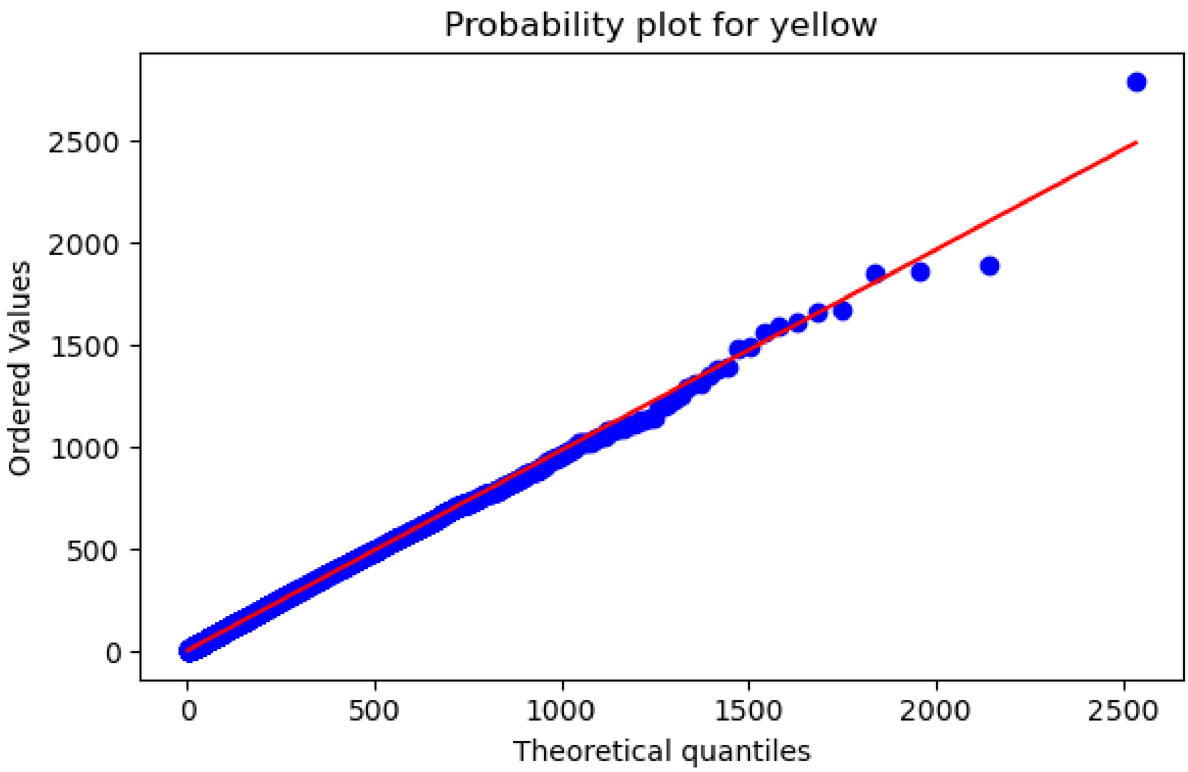
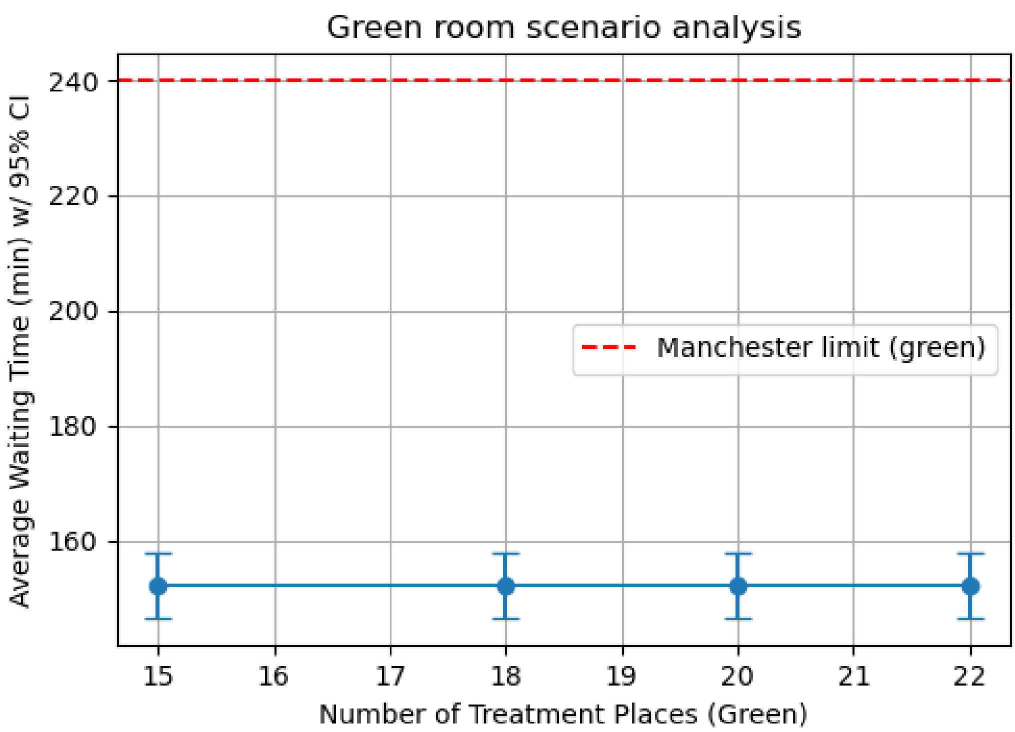


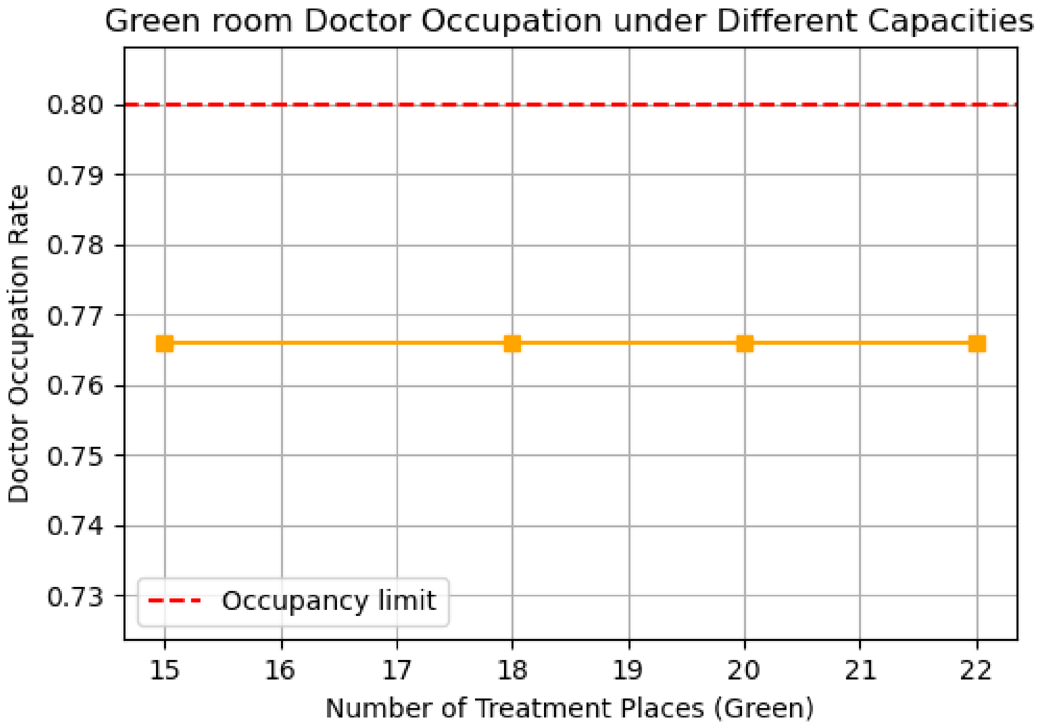
Figure 6: Treatment time distribution for Yellow priority patients and corresponding probability plot. The data are pooled from multiple runs to achieve stable parameter estimates. While the fit is generally good, a few extreme outliers underscore the need for sufficient treatment capacity to handle complex cases.

Scenario Analyses

Green Room Scenario

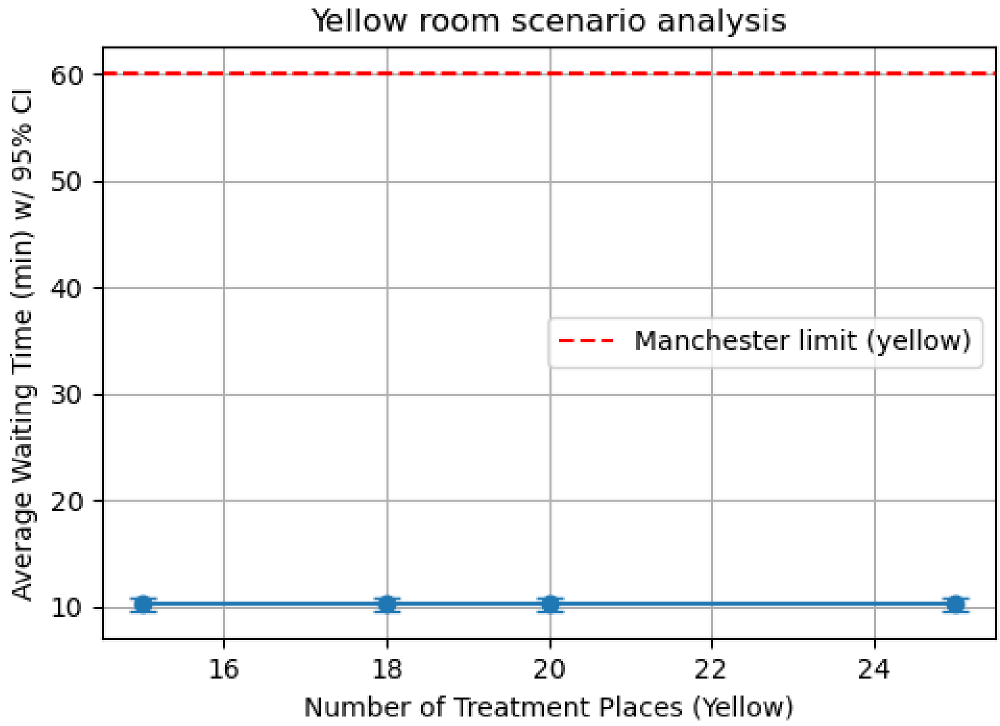
Green room scenarios varied treatment capacities between 15 and 22 places, showing that even at reduced capacities, waiting times remained well below the 240-minute limit. **Figure 7** demonstrates the mean waiting time under different capacities. Doctor occupation rates in these scenarios also stayed under the critical threshold, as shown in **Figure 8**, indicating balanced utilization.

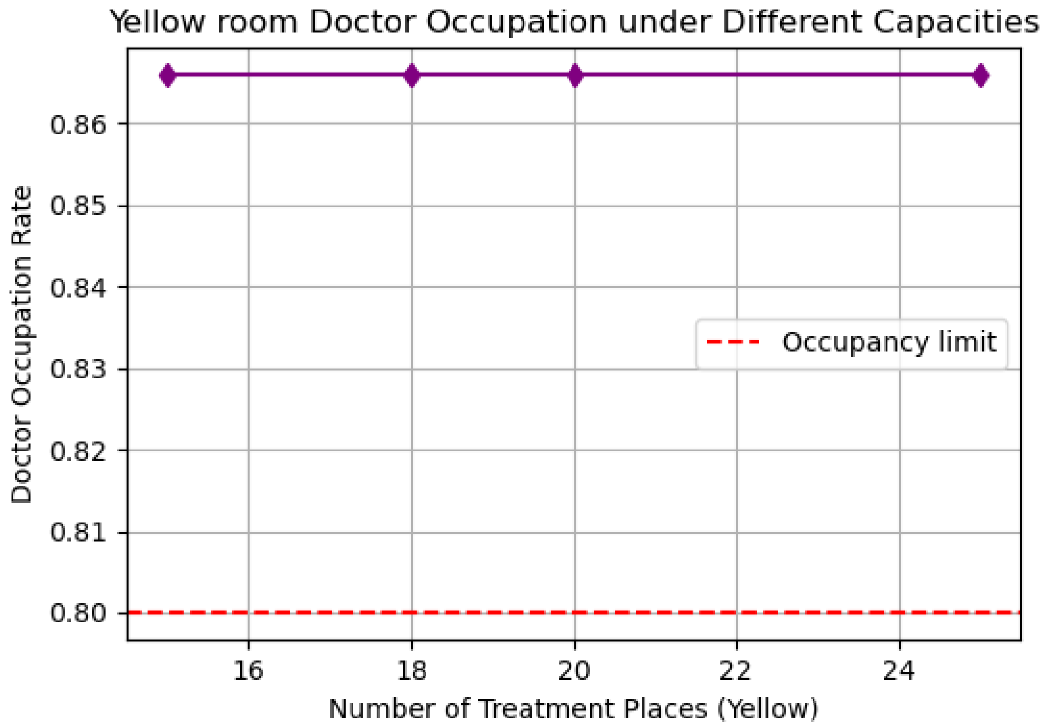




Yellow Room Scenario

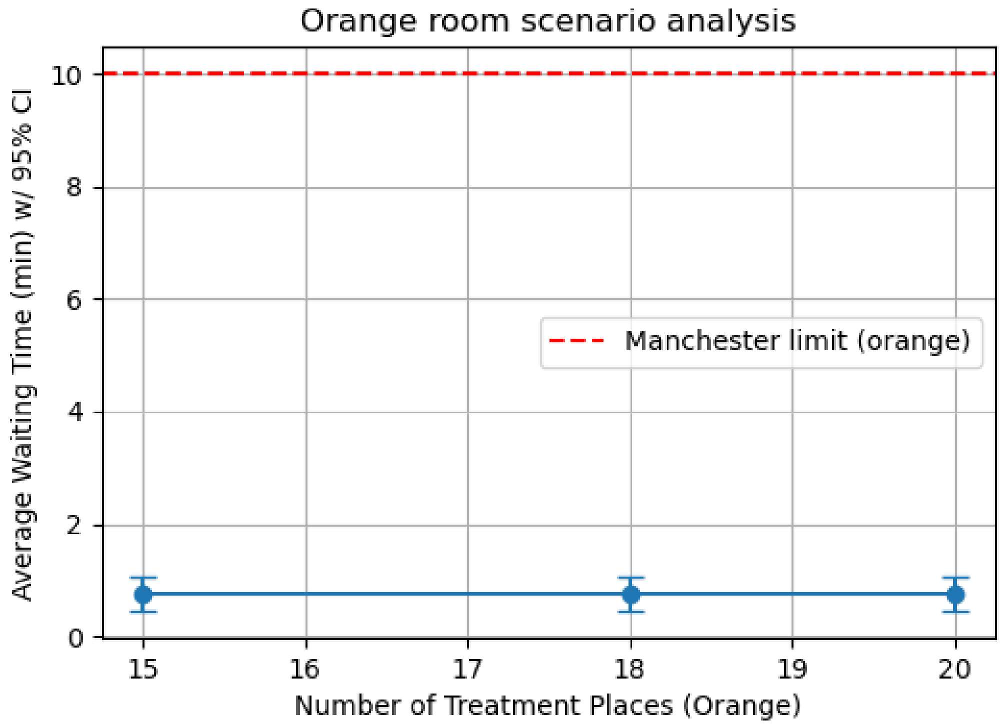
Yellow room scenarios involved testing capacities from 15 to 25 places. Waiting times consistently remained below the 60-minute Manchester limit, as shown in **Figure 9**. Doctor occupation rates were higher than in the Green room but stayed within manageable levels, ensuring no overload. **Figure 10** illustrates the occupation rates for these scenarios.

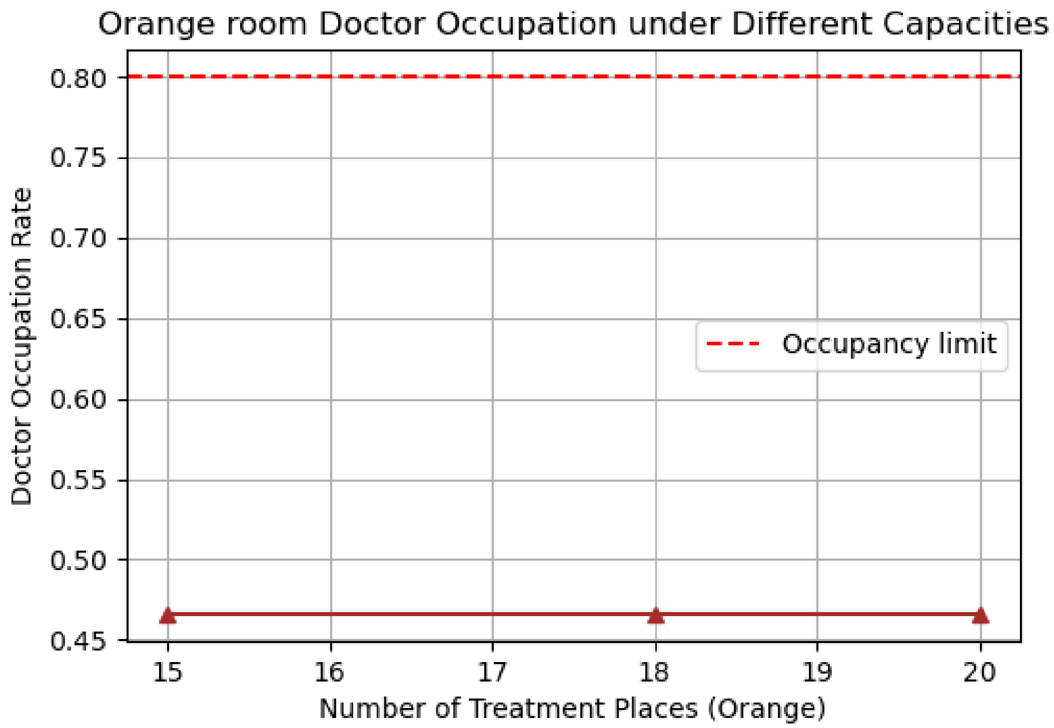




Orange Room Scenario

In the Orange room scenarios, where the Manchester limit is 10 minutes, all tested capacities maintained mean waiting times near zero. **Figure 11** presents the waiting times across scenarios. Doctor occupation rates were also low, suggesting potential over-resourcing or efficient system design, as seen in **Figure 12**.





Discussion

- **Arrival and Admission:** The aggregated hourly arrival rates highlight the importance of dynamic staffing. Admission times are short and stable, indicating this step is not a major bottleneck.

- **Triage and Treatment Time Distributions:** Fitting a lognormal distribution to triage and Yellow treatment times reveals a skewed profile, with most patients processed quickly but a tail of slower cases. This justifies providing a small buffer in resources to handle outliers without affecting overall system performance.
- **Scenario Analyses:**
    - **Green Room:** Reducing treatment capacity does not significantly impact waiting times or push doctor occupation to critical levels, suggesting current configurations might be optimized for cost savings without risking delays.
    - **Yellow Room:** Increasing beds maintains low waiting times and manageable doctor occupation, reflecting a good match between capacity and demand.
    - **Orange Room:** With very tight Manchester limits, the system still easily meets target times. The low doctor occupation rates suggest the possibility of reallocating resources if needed, although ensuring immediate care for Orange priority cases remains paramount.

Each scenario result is based on averages from at least five independent simulation runs (using distinct random seeds) to ensure statistical reliability. We also provide 95% confidence intervals to quantify uncertainty. Across all scenarios and priorities, waiting times remain below the respective Manchester limits, and resource occupation stays within acceptable bounds.

Overall, these results demonstrate a well-performing system across all tested configurations. Key metrics like waiting times and resource utilization remained within acceptable bounds, validating the robustness of the simulation model. Statistical rigor was maintained through multiple runs and confidence intervals, ensuring reliable insights into optimizing emergency department operations.

Conclusion

Summary of Simulation Results

This simulation study quantitatively assessed the impact of various resource configurations and patient arrival dynamics on key performance metrics, including waiting times, treatment durations, and resource utilization. By conducting multiple independent simulation runs (at least five per scenario) with different initial random seeds, we obtained statistically stable estimates and 95% confidence intervals for all reported results.

Key quantitative findings are:

- **Arrival Patterns & Admission:** The aggregated arrival rate data revealed distinct daily and hourly peaks. Despite these fluctuations, admission times remained short and stable (typically between 3 to 5 minutes), indicating that the initial patient intake process is not a major bottleneck.
- **Triage & Treatment Distributions:** Triage times and Yellow patient treatment times followed a lognormal distribution with a prominent long tail. While the majority of patients were processed efficiently, a subset required substantially longer times. These outliers highlight the importance of maintaining a buffer in staffing and resources to accommodate uncommon but demanding cases without compromising overall service levels.
- **Scenario Analyses (Green, Yellow, Orange):**
  - **Green Zone:** Even with reduced treatment capacity (e.g., from 22 to 15 treatment places), average waiting times remained well under the Manchester target of 240 minutes. Doctor occupation rates also stayed moderate, suggesting that the current Green zone configuration is robust and potentially over-resourced.
  - **Yellow Zone:** Increasing the number of treatment places helped maintain waiting times comfortably below the 60-minute Manchester limit. Although doctor occupation rates were higher than in the Green scenario, they remained within acceptable bounds, balancing efficient patient throughput with manageable staff workload.
  - **Orange Zone:** With the strictest Manchester limit of 10 minutes, the system consistently delivered rapid treatment. Across tested capacities, waiting times were nearly negligible, and doctor occupation rates were notably low, indicating highly efficient handling of the most urgent cases.

Overall, all examined configurations met the respective Manchester standards, demonstrating a well-performing system. The data-driven insights confirm that current resource allocations maintain patient flow without overburdening clinical staff.

Conclusions Based on the Simulation Study

1. **Resource Robustness:** The system, as configured, appears resilient across multiple scenarios. Even under lower capacities (Green zone) or higher demands (Yellow zone), it maintains waiting times below critical thresholds and avoids staff overload.
2. **Importance of Flexibility:** Although average times are comfortably within targets, the presence of long-tail distributions (for triage and treatment) underscores the

necessity for some resource flexibility. The system should be capable of handling occasional surges or complex cases without significant performance degradation.

3. **Prioritization Works:** The differentiation into Green, Yellow, and Orange priority zones ensures that the most critical patients (Orange) receive near-immediate attention, while less urgent patients still experience acceptable waiting times. This tiered approach leverages resources efficiently and meets established clinical standards.

Recommendations for Improvement

- **Data-Driven Scheduling:** Since arrival patterns vary throughout the day and week, dynamic staffing schedules could further optimize resource utilization. Increasing staff during known peak hours and reducing them during low-demand periods may achieve cost savings without impacting patient care.
- **Monitoring Long-Tail Cases:** Consider strategies (e.g., adding a floating nurse or physician) to quickly respond when an unusually long treatment or triage case arises. Small resource adjustments could mitigate the impact of outlier cases.
- **Scenario Expansion and Sensitivity Analysis:** Future studies could test even more diverse configurations (e.g., different staff-to-bed ratios, altered triage protocols, or changing patient arrival distributions). Such sensitivity analyses would help identify thresholds at which system performance significantly declines, providing insights for targeted improvements.

In conclusion, this simulation-based study highlights a stable, flexible system that meets established performance targets across various scenarios. By using multiple runs, confidence intervals, and scenario-based analyses, we have identified strengths in the current configuration and provided actionable suggestions to enhance efficiency and resilience in future implementations.

References:

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